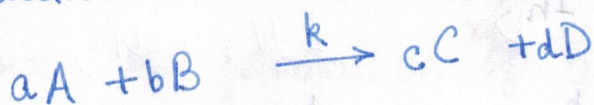


Chemical Kinetics

(11)

- * Concerns itself with the measurement of rate of reactions proceeding under given conditions of temperature, pressure and concentration.
- * no correlation b/w thermodynamic instability and the rate of a chemical reaction.

* Rate of reaction



$$\text{Reaction rate, } r = -\frac{1}{a} \frac{d[A]}{dt} = -\frac{1}{b} \frac{d[B]}{dt} = \frac{1}{c} \frac{d[C]}{dt} = \frac{1}{d} \frac{d[D]}{dt} \quad \text{--- (1)}$$

$$\text{Rate law expression, } r = k[A]^a[B]^b \quad \text{--- (2)}$$

where k = rate constant (defined as rate of reaction when the concentration of each reactant is unity.)

* Order of the Reaction $\Rightarrow$ defined as the sum of the ~~poor~~ powers to which the concentration (or pressure) terms are raised in the rate law expression.

for equation (2) ^{order} of the reaction = $a + b$

* for an n -th order reaction, the units of $k_n = (\text{dm}^3)^{n-1} \text{mol}^{1-n} \text{s}^{-1}$

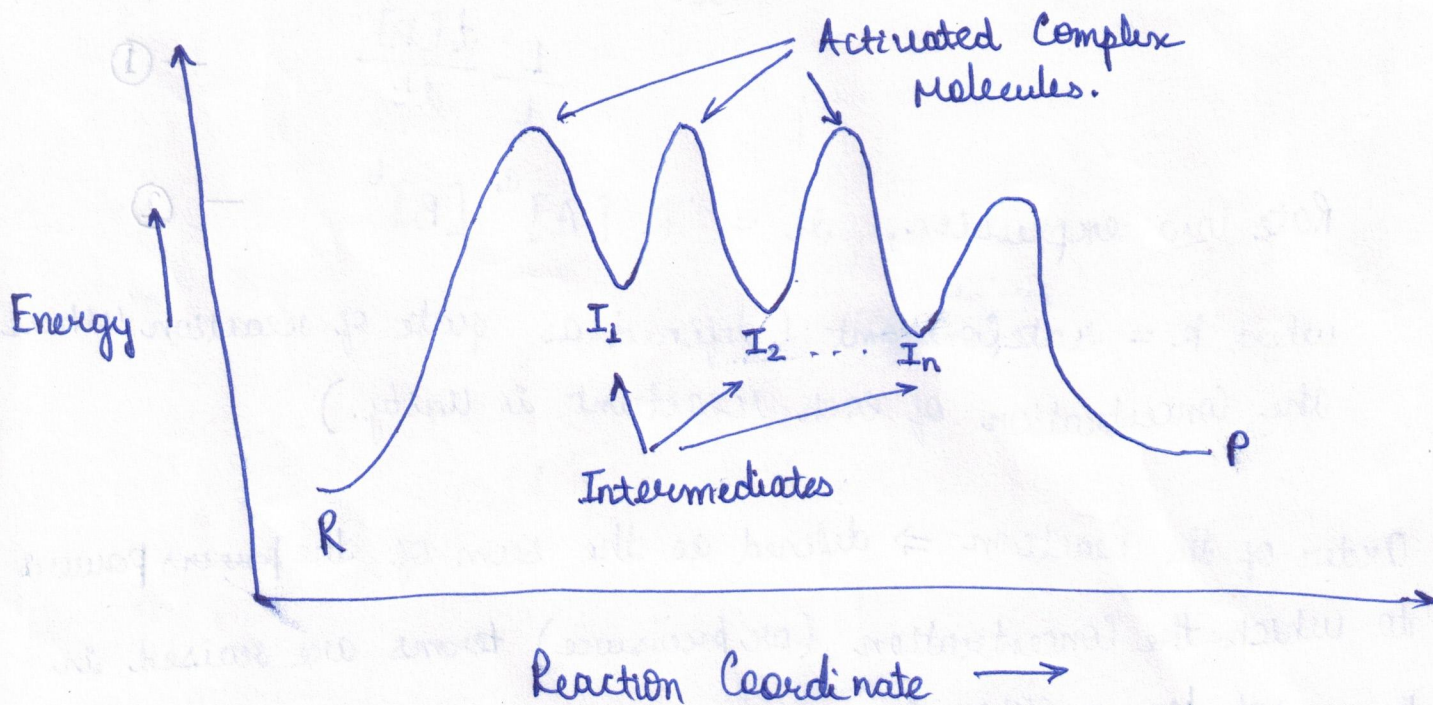
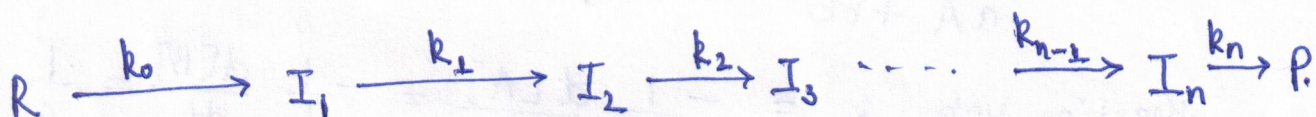
* Molecularity of a reaction $\rightarrow$ defined as the number of molecules involved in the step leading to the chemical reaction.

* Mechanisms of Complex Reactions.

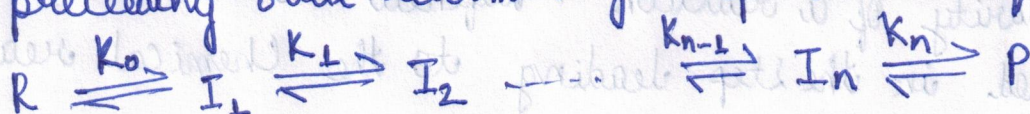
Two approximations are generally used for elucidating the mechanism of the complex reaction:

① equilibrium approximation

Consider a reaction in which reactant R gives rise to product P through the formation of a series of consecutive intermediates $I_1, I_2 \dots I_n$



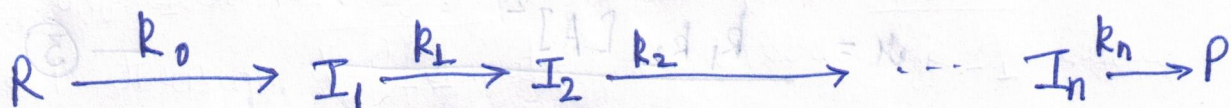
In order to derive the rate equation for the reaction, $\rightarrow$
 Assume a rate determining step exists which is the slowest
 in a sequence of steps. It is further assumed that all
 steps preceding rate determining steps are in equilibrium



where K_i 's are the various equilibrium constants.

⑤ The Steady State

In a case where the reactions are investigated under such conditions that the slowest rate determining ~~step~~ step does not exist, one assumes the steady state approximation for the short lived intermediates.

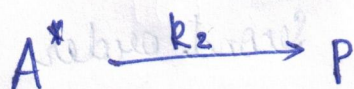
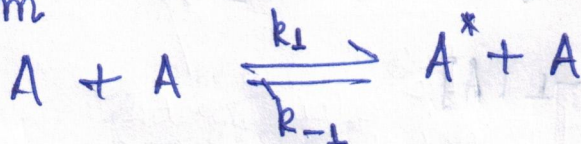


$\therefore$ the rate of formation of an intermediate = rate of its decomposition.

$$\Rightarrow \frac{d[I_1]}{dt} = \frac{d[I_2]}{dt} = \dots = \frac{d[I_n]}{dt} = 0$$

⑥ Lindemann theory of unimolecular reactions.

Consider the unimolecular reaction $A \rightarrow P$ which proceeds via mechanism



$A^* \rightarrow$ is the ~~energ~~ energized A molecule which has acquired sufficient vibrational energy to enable it to isomerize or decompose.

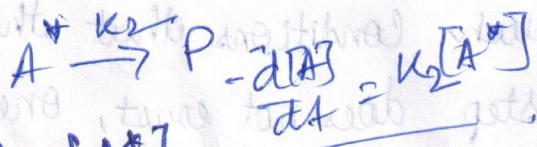
$A^* \rightarrow$ is simply a molecule in a high vibrational energy level and not an activated complex.

According to steady state approximation,

$$\frac{d[A^*]}{dt} = 0 = k_1[A]^2 - k_{-1}[A][A^*] - k_2[A^*]$$

$$\Rightarrow [A^*] = \frac{k_1 [A]^2}{k_{-1} [A] + k_2}$$

Rate of reaction is given by



$$r = -\frac{d[A]}{dt} = k_2 [A^*]$$

$$r = \frac{k_1 k_2 [A]^2}{k_{-1} [A] + k_2} \quad \text{--- (3)}$$

two limiting cases:

Case 1: if $k_{-1} [A] \gg k_2$

$$\Rightarrow r = \left(\frac{k_1 k_2}{k_{-1}} \right) [A] \quad \text{first order reaction.}$$

In a gaseous reaction, this is the high pressure limit because at very high pressure $[A]$ is very large $\Rightarrow k_1 [A] \gg k_2$

Case 2: if $k_2 \gg k_{-1} [A]$

$$\Rightarrow r = k_1 [A]^2 \quad \text{Second order reaction}$$

low pressure limit.

(*) The experimental rate is defined as.

$$R = k_{uni} [A]$$

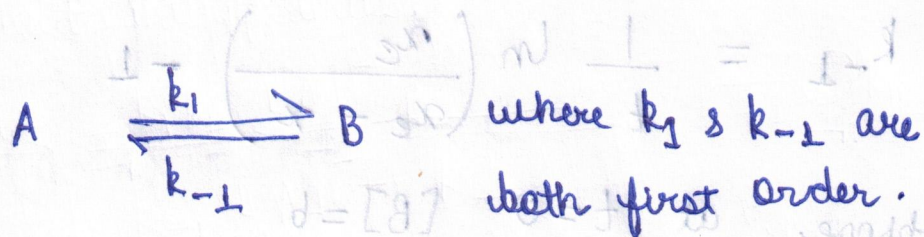
comparing with eq (3)

$$\Rightarrow k_{uni} = \frac{k_1 k_2}{k_{-1} + k_2 / [A]}$$

$$[A] \rightleftharpoons [A^*] \quad [A]_{eq} = [A^*]_{eq} = 0 = \frac{[A]_{eq}}{+b}$$

* Opposing or reversible Reactions.

Kinetics of such rxn are usually studied in the initial stages of the process when the products are at too low a conc to set up the opposing reaction at a noticeable rate.



$$\text{at } t=0 \quad [A] = a, \quad [B] = 0$$

$$\text{at } t=t \quad [A] = a-x, \quad [B] = x$$

$$\text{net rate of formation of B} \quad \frac{dx}{dt} = k_1(a-x) - k_{-1}x$$

if x_e is the equilibrium conc of B (ie when the net rate = 0)

$$k_1(a-x_e) - k_{-1}x_e = 0$$

$$\Rightarrow k_{-1} = k_1(a-x_e)/x_e$$

$$\Rightarrow \frac{dx}{dt} = k_1 \frac{(x_e - x)a}{x_e}$$

$$\Rightarrow \int \frac{dx}{(x_e - x)} = \int \frac{k_1 a}{x_e} dt$$

$$\Rightarrow -\ln(x_e - x) = \frac{a}{x_e} (k_1 t + C)$$

where C is constant of integration.

at $t=0$, $x=0$

$$\Rightarrow C = -\frac{x_e}{a} \ln x_e$$

$$\Rightarrow k_1 = \frac{x_e}{at} \ln \left(\frac{x_e}{x_e - x} \right)$$

$$k_{-1} = \frac{1}{t} \ln \left(\frac{x_e}{x_e - x} \right) - 1$$

⊗ Suppose at $t=0$ $[B]=b$
net rate of formation of B, $\frac{dx}{dt} = k_1(a-x) - k_{-1}(b+x)$

$$\Rightarrow k_{-1} = k_1(a-x_e) / (b+x_e)$$

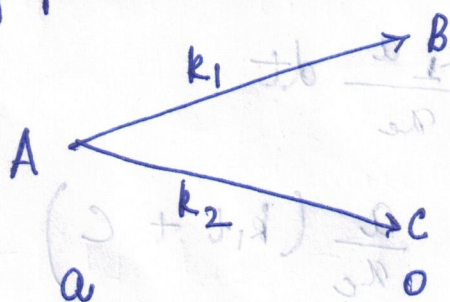
$$\Rightarrow \frac{dx}{dt} = k_1 \left(\frac{a+b}{b+x_e} \right) (x_e - x)$$

after integrating,

$$k_1 = \frac{b+x_e}{t(a+b)} \ln \left(\frac{x_e}{x_e - x} \right)$$

$$\Delta (k_1 + k_{-1}) = \frac{1}{t} \ln \left(\frac{x_e}{x_e - x} \right)$$

⊗ kinetics of parallel reaction.



at $t=0$

at $t=t$

a

a-x

0

x

Net Rate of reaction.

$$\frac{dx}{dt} = k_1(a-x) + k_2(a-x)$$
$$= K(a-x) \quad \text{where } K = k_1 + k_2$$

$$\Rightarrow \int \frac{dx}{(a-x)} = \int K dt$$

$$\Rightarrow -\ln(a-x) = Kt + C$$

at $t=0$, $x=0$

$$\Rightarrow C = -\ln a$$

$$\Rightarrow K = \frac{1}{t} \ln \left(\frac{a}{a-x} \right)$$

$$\Rightarrow \text{let } \frac{k_1}{k_2} = K'$$

$$\Rightarrow k_1 = K' \cdot k_2$$

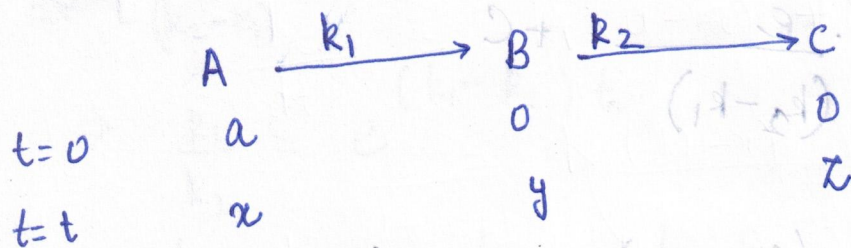
$$\therefore k_1 + k_2 = K$$

$$\Rightarrow K = K' \cdot k_2 + k_2$$

$$\Rightarrow k_2 = \frac{K}{(1 + K')}$$

$$k_1 = \frac{K K'}{(1 + K')}$$

⊗ Consecutive Reaction.



where k_1, k_2 are rate constants, & $k_1 \neq k_2$

$$a = x + y + z$$

Rate of disappearance of A:

$$-\frac{dx}{dt} = k_1 x$$

⇒ integrating both sides and computing the value of integration constant.

$$\frac{x}{a} = e^{-k_1 t} \Rightarrow x = a e^{-k_1 t} \quad (1)$$

Rate of formation of C:

$$\frac{dz}{dt} = k_2 y \quad (2)$$

Rate of accumulation of B:

$$\frac{dy}{dt} = -\frac{dx}{dt} - \frac{dz}{dt} = k_1 x - k_2 y = k_1 a e^{-k_1 t} - k_2 y$$

$$\Rightarrow \frac{dy}{dt} + k_2 y = k_1 a e^{-k_1 t} \quad (3)$$

multiplying both sides by $e^{k_2 t}$

$$\left(\frac{dy}{dt} + k_2 y \right) e^{k_2 t} = k_1 a e^{-(k_1 - k_2)t}$$

$$\Rightarrow d(y \cdot e^{k_2 t}) = k_1 a e^{-(k_2 - k_1)t} dt$$

integrating both sides

$$y \cdot e^{k_2 t} = k_1 a \cdot \frac{e^{(k_2 - k_1)t}}{(k_2 - k_1)} + C$$

at $t=0, y=0$

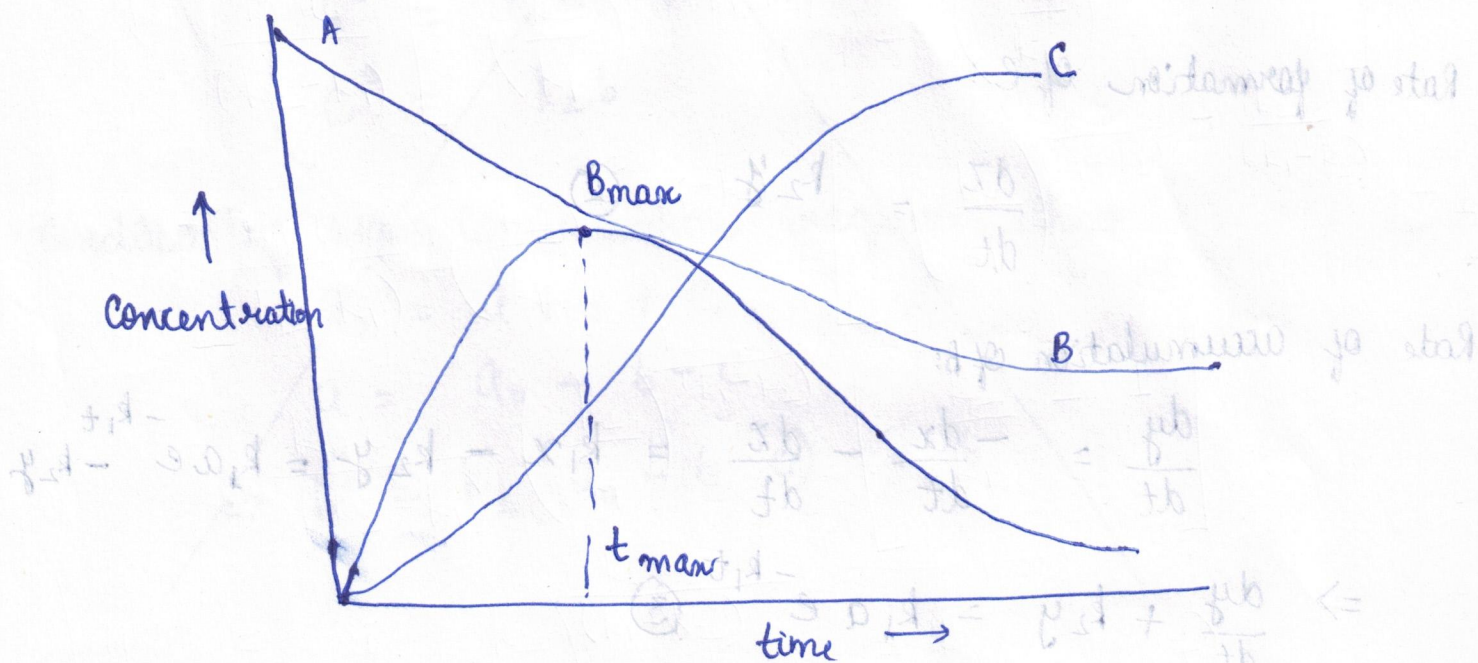
$$\Rightarrow C = -k_1 a / (k_2 - k_1)$$

$$\Rightarrow y = \frac{k_1 a}{(k_2 - k_1)} \left[e^{-k_1 t} - e^{-k_2 t} \right] \quad (4)$$

since $a = x + y + z$

$$z = a - x - y$$

$$\Rightarrow z = a \left[(k_2 - k_2 e^{-k_1 t}) - (k_1 - k_1 e^{-k_2 t}) \right] \quad (5)$$



⊛ As the reaction proceeds [B] reaches a maximum and falls off thereafter

for $t_{max}, \frac{dy}{dt} = 0$

$$\frac{d}{dt} \left[\frac{ak_1}{(k_2 - k_1)} (e^{-k_1 t} - e^{-k_2 t}) \right] = 0$$

$$\Rightarrow \frac{k_1}{k_2} = e^{(k_1 - k_2)t}$$

$$\Rightarrow t_{\max} = \frac{1}{(k_1 - k_2)} \ln \frac{k_2}{k_1} \quad (6)$$

from eq (4)

$$y = \frac{k_1 a}{(k_2 - k_1)} \left[e^{-k_1 t} - e^{-k_2 t} \right]$$

putting $t_{\max}$ value in above equation

$$y = \frac{k_1 a}{(k_2 - k_1)} \left[\left(\frac{k_2}{k_1} \right)^{k_1/(k_1 - k_2)} - \left(\frac{k_2}{k_1} \right)^{k_2/(k_1 - k_2)} \right]$$

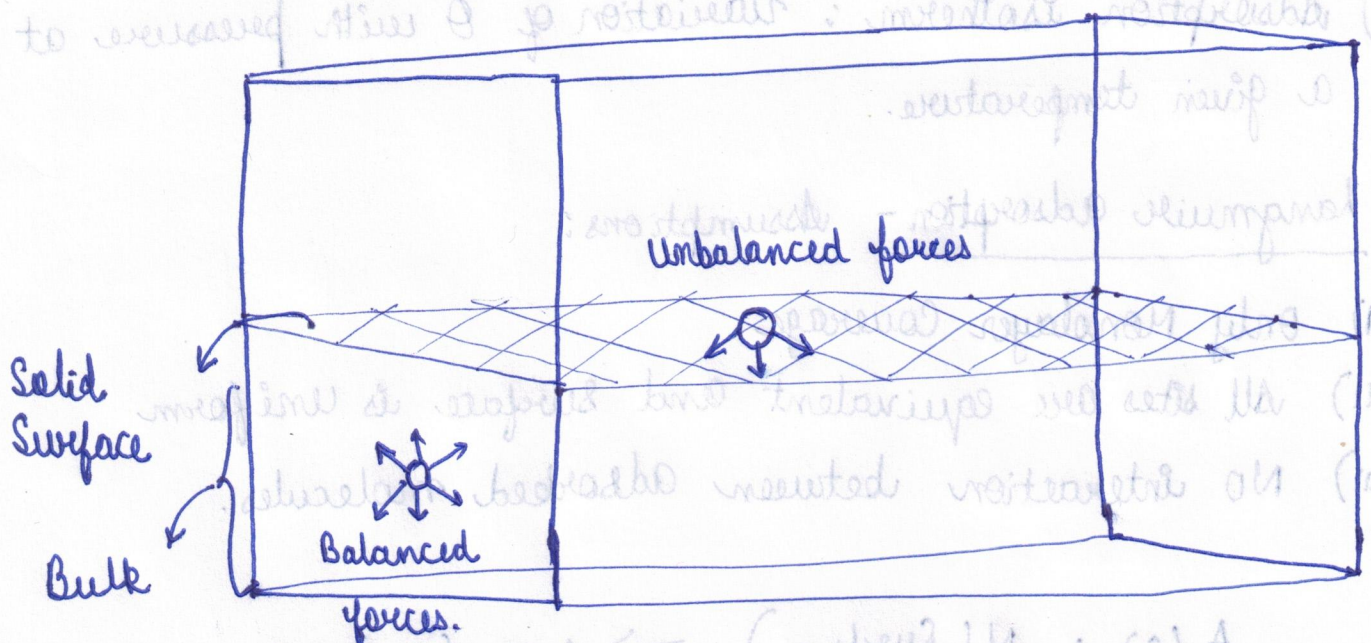
$$= \frac{k_1 a}{(k_2 - k_1)} \cdot \left(\frac{k_2}{k_1} \right)^{k_2/(k_1 - k_2)} \left[\left(\frac{k_2}{k_1} \right)^{\frac{k_1}{(k_1 - k_2)}} - \frac{k_2}{(k_1 - k_2)} - 1 \right]$$

$$= \frac{k_1 a}{(k_2 - k_1)} \left(\frac{k_2}{k_1} \right)^{\frac{k_2}{(k_1 - k_2)}} \left[\frac{k_2}{k_1} - 1 \right]$$

$$y = a \cdot \left(\frac{k_2}{k_1} \right)^{\frac{k_2}{(k_1 - k_2)}} = B_{\max} \quad (7)$$

①
* Adsorption: Adsorption is a mass transfer process that is a phenomenon of gases or solutes by solid or liquid surfaces.

- > Molecules or atoms on the solid surfaces have residual surface energy due to unbalanced forces. When some substance collide with the solid surface, they are attracted by these unbalanced forces and stay on the solid surface.



* Fractional Coverage $\theta = \frac{\text{number of adsorption sites occupied}}{\text{number of adsorption sites available.}}$

- > θ - Often expressed in terms of the volume of adsorbate adsorbed by $\theta = V/V_{\infty}$ where V_{∞} is the volume of adsorbate corresponding to complete monolayer coverage.

* Types of Adsorption

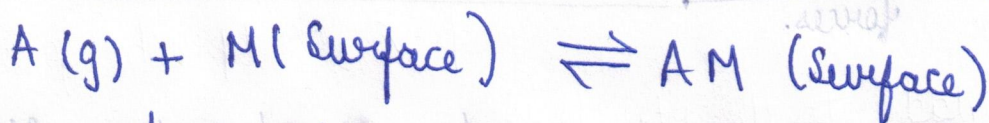
i) physisorption - van der Waals interaction b/w adsorbate and the substrate.

ii) Chemisorption - molecules stick to the surface by forming a chemical (usually covalent) bond, and tend to find sites that maximizes their coordination number with the substrate.

* Adsorption isotherm: variation of θ with pressure at a given temperature.

* Langmuir Adsorption - assumptions:

- i) Only Monolayer Coverage
- ii) All sites are equivalent and surface is uniform
- iii) No interaction between adsorbed molecules.



forward rate constant (for adsorption) = k_a

reverse rate constant (for desorption) = k_d

Rate of adsorption $\frac{d\theta}{dt} = k_a p N (1 - \theta)$

N = total number of sites

p = Gas's partial pressure.

Rate of desorption $\frac{d\theta}{dt} = -k_d N \theta$

At equilibrium, there is no net change. (3)

$$k_a p N (1 - \theta) = -k_d N \theta$$

$$\Rightarrow \theta = \frac{K p}{1 + K p}$$

$$K = \frac{k_a}{k_d}$$

Langmuir isotherm

$$\Rightarrow K p \theta + \theta = K p$$

$$\therefore \theta = V/V_{\infty}$$

$$\frac{p}{V} = \frac{p}{V_{\infty}} + \frac{1}{K V_{\infty}}$$

$\Rightarrow$ plot of p/V vs $p \rightarrow$ gives a straight line of slope $1/V_{\infty}$ and intercept $1/K V_{\infty}$

(*) for adsorption with dissociation,

rate of adsorption $\propto p$

$\propto$ probability that both atoms will find sites $\propto$ square of number of vacant sites.

$$\frac{d\theta}{dt} = k_a p \{ N (1 - \theta) \}^2$$

Rate of desorption $\propto$ frequency of encounters of atoms on the surface

$$\frac{d\theta}{dt} = -k_d (N\theta)^2 \quad \text{at equilibrium, } \frac{d\theta}{dt} = 0 \quad (4)$$

at equilibrium,

Langmuir isotherm

$$\theta = \frac{(K_p)^{1/2}}{1 + (K_p)^{1/2}}$$

Langmuir isotherm for adsorption with dissociation

$$q = \theta + \theta q \Rightarrow$$

$$\frac{q}{V} = \theta \quad \therefore$$

$$\frac{1}{K_p V} + \frac{q}{V} = \frac{q}{V}$$

plot of q/V vs $1/V$ is a straight line

intercept $1/K_p$ and slope $1/V$

for adsorption with dissociation (*)

rate of adsorption & rate of desorption

are equal at equilibrium

will give a straight line

number of vacant sites

$$\frac{d\theta}{dt} = k_a p (1-\theta) - k_d \theta^2$$

rate of adsorption & rate of desorption are equal at equilibrium

Assignment 1

- ① At 1100 K the following data were obtained on the homogeneous reaction:



	[NO] (mol dm ⁻³)	[H ₂] (mol dm ⁻³)	Rate, r (mol dm ⁻³ s ⁻¹)
i)	5×10^{-3}	2.5×10^{-3}	3×10^{-5}
ii)	15×10^{-3}	2.5×10^{-3}	9×10^{-5}
iii)	15×10^{-3}	10.0×10^{-3}	36×10^{-5}

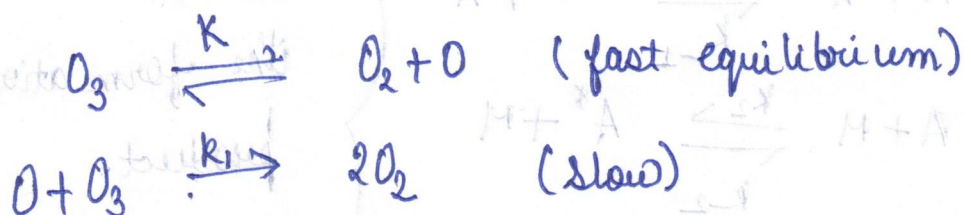
- ① Calculate the order of the reaction with respect to NO, H₂ and the overall reaction order.

- ② Write the rate law expression for the reaction.

- ③ Calculate the rate constant of the reaction.

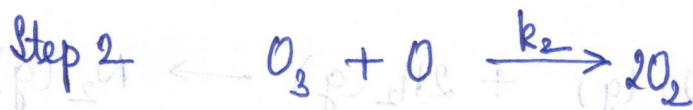
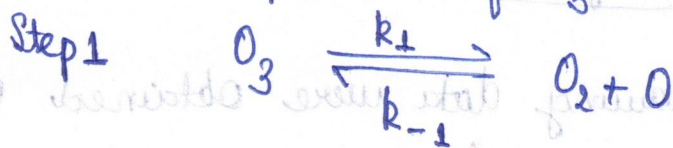
- ④ What will be the rate of reaction if $[\text{NO}] = [\text{H}_2] = 8 \times 10^{-3} \text{ mol/dm}^3$.

- ② The gaseous decomposition of O_3 , $2\text{O}_3 \longrightarrow 3\text{O}_2$ obeys the rate law $r = -d[\text{O}_3]/dt = k[\text{O}_3]^2/[\text{O}_2]$. Show that the following mechanism is consistent with the above rate law:



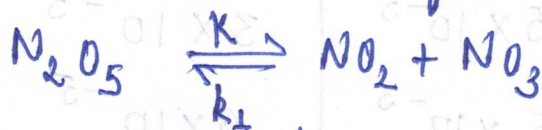
③ for the thermal decomposition of O_3 to $O_2 \rightarrow$

(42)

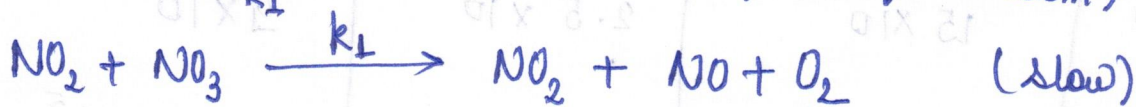


what is the observed rate law?

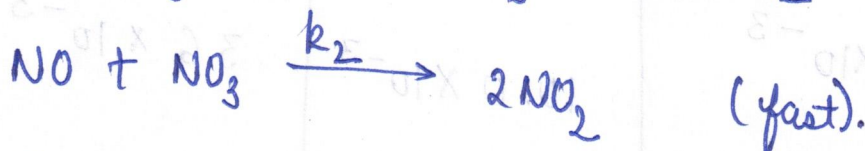
④ what is the rate law for $N_2O_5 \longrightarrow 2NO_2 + \frac{1}{2}O_2$?



(fast equilibrium)

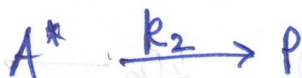
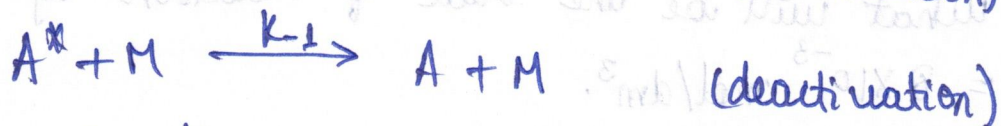
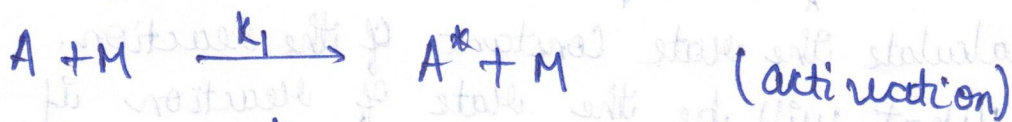


(slow)

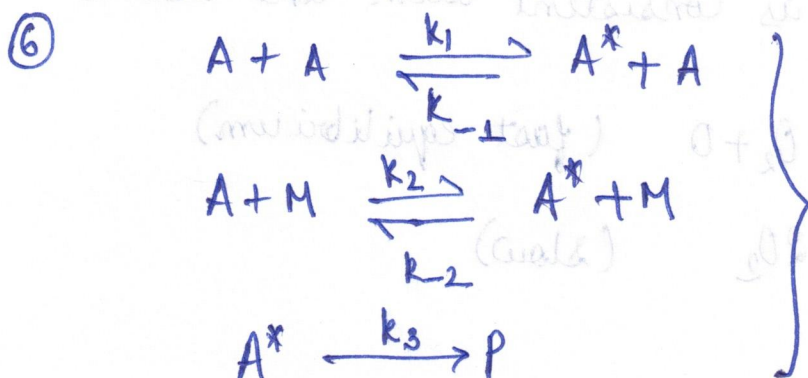


(fast).

⑤ Consider the following Lindemann mechanism for the unimolecular decomposition of a molecule A in the presence of a species M



Derive the rate law for the formation of the product.



Derive the rate law for the formation of the product.

$$\textcircled{1} \quad r = k [\text{NO}]^a [\text{H}_2]^b$$

$$3 \times [\text{NO}] \text{ with } [\text{H}_2] \text{ constant} \rightarrow 3 \times r \Rightarrow a = 1$$

$$4 \times [\text{H}_2] \text{ with } [\text{NO}] \text{ constant} \rightarrow 4 \times r \Rightarrow b = 1$$

$$\Rightarrow r = k [\text{NO}] [\text{H}_2]$$

$$\textcircled{ii} \quad k = \frac{r}{[\text{NO}] [\text{H}_2]} = 2.48 \text{ dm}^3 / \text{mol} \cdot \text{s}$$

(for any set of the data).

$$\textcircled{iv} \quad r = k [\text{NO}] [\text{H}_2] = 1.536 \times 10^{-4} \text{ mol} / \text{dm}^3 \cdot \text{s}$$

② for slow rate determining step

$$r = - \frac{d[\text{O}_3]}{dt} = k_1 [\text{O}] [\text{O}_3]$$

from fast equilibrium step

$$K = \frac{[\text{O}_2] [\text{O}]}{[\text{O}_3]}$$

$$\Rightarrow [\text{O}] = K [\text{O}_3] / [\text{O}_2]$$

$$\Rightarrow r = k_1 K [\text{O}_3]^2 / [\text{O}_2] = k [\text{O}_3]^2 / [\text{O}_2]$$

where $k = k_1 K$.

$$\textcircled{3} \quad r = \frac{d[\text{O}_3]}{dt} = -k_1 [\text{O}_3] + k_{-1} [\text{O}_2] [\text{O}] - k_2 [\text{O}_3] [\text{O}]$$

using steady state approx for $[\text{O}]$

$$\frac{d[\text{O}]}{dt} = k_1 [\text{O}_3] - k_{-1} [\text{O}_2] [\text{O}] - k_2 [\text{O}_3] [\text{O}] = 0$$

$$\Rightarrow [\text{O}] = k_1 [\text{O}_3] / (k_{-1} [\text{O}_2] + k_2 [\text{O}_3])$$

$$\Rightarrow r = \frac{-2 k_1 k_2 [O_3]^2}{k_{-1} [O_2] + k_2 [O_3]}$$

assume $k_{-1} [O_2] \gg k_2 [O_3]$

$$\Rightarrow r = -k \frac{[O_3]^2}{[O_2]} \quad \text{where } k = \frac{2 k_1 k_2}{k_{-1}}$$

④ $r = k_1 [NO_2] [NO_3]$

from fast equilibrium step

$$K = \frac{[NO_2] [NO_3]}{[N_2O_5]}$$

$$\Rightarrow [NO_2] [NO_3] = K [N_2O_5]$$

$$\Rightarrow r = k_1 K [N_2O_5] = k [N_2O_5]$$

where $k = k_1 K$.

⑤ $r = -\frac{d[A]}{dt} = \frac{d[P]}{dt} = k_2 [A^*]$

using steady state approx

$$k_1 [A] [M] - k_{-1} [A^*] [M] - k_2 [A^*] = 0$$

$$[A^*] = \frac{k_1 [A] [M]}{k_{-1} [M] + k_2}$$

$$\Rightarrow r = \frac{k_1 k_2 [M] [A]}{k_{-1} [M] + k_2}$$

$$k_{-1} [M] + k_2$$

⑥

$$u = k_3 [A^*]$$

using steady state approx

$$k_1 [A]^2 - k_{-1} [A^*] [A] + k_2 [A] [M] - k_{-2} [A^*] [M] - k_3 [A^*] = 0$$

$$[A^*] = \frac{k_1 [A]^2 + k_2 [A] [M]}{k_{-1} [A] + k_{-2} [M] + k_3}$$

$$\Rightarrow u = k_3 \left(\frac{k_1 [A]^2 + k_2 [A] [M]}{k_{-1} [A] + k_{-2} [M] + k_3} \right)$$